

MUHAMED AMIN HASSAN

Flutter & Android Developer

Cairo, Egypt | muhamed.amminn@gmail.com | +201027660252 | Military: Completed

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Flutter & Android Developer with hands-on experience building and maintaining production mobile applications, including Flutter apps integrated with Odoo ERP and native Android applications. Experienced with Flutter, Kotlin, Clean Architecture, BLoC/Cubit, Riverpod, Firebase, REST/JSON-RPC APIs, and Python/Odoo backend development. Built AI-powered and offline-first personal projects demonstrating experience in AI integration, real-time systems, local persistence, testing, and CI/CD.

WORK EXPERIENCE

Flutter Developer Intern — [Knowledge BI](#) — [Odoo Gold Partner](#) Jul 2026 – Present

Working on two production Flutter applications integrated with Odoo ERP, serving real clients in agriculture and HR sectors:

- Developing and maintaining AI-Asal El-Barry (HR/Attendance) and Sahatk Falahy (E-Commerce) — both live on Google Play and the App Store with 10,000+ combined downloads.
- Building features across the full stack: Flutter UI/UX with Riverpod state management, RESTful API integration using the dio package, and Odoo backend endpoint development (Python/Odoo ORM) for attendance, authentication, and e-commerce workflows.
- Led Android release pipeline setup for Sahatk Falahy: keystore generation, Gradle signing configuration (Kotlin DSL), and Play Store submission preparation.
- Resolved complex e-commerce bugs across address management, location/zone selection, and Odoo-driven delivery pricing logic.
- Collaborated with the ERP integration team to design and consume Odoo JSON-RPC endpoints, bridging mobile features with server-side business logic.

Android Developer Intern — [Zimozi Solutions](#) Aug 2022 – Feb 2023

- Delivered 2 production Android applications (Evegene Quiz, Claim Exchange) in Kotlin/MVVM, reducing code complexity by 30% through Clean Architecture modular decomposition.
- Implemented Dagger Hilt DI and engineered 10+ reusable Jetpack Compose components, cutting feature development time by ~25%.
- Optimized 5+ REST API integrations with Retrofit, reducing API response processing overhead by up to 35%.
- Contributed Kotlin/JS components for cross-platform Web & Mobile synchronization, eliminating 40% of UI inconsistencies across platforms.

PRODUCTION PROJECTS

AI-Asal El-Barry — HR & Attendance Management App

Stack: Flutter · Dart · Riverpod · Odoo JSON-RPC · http · SharedPreferences · Android · iOS

Available on: [Google Play](#) | [App Store](#)

- Production HR/attendance app serving agricultural workforce management, live on both platforms with 2,000+ downloads.
- Implemented a robust check-in/check-out attendance system with GPS-aware state tracking and persistent session management.
- Developed and consumed custom Odoo backend endpoints (Python/Odoo ORM) for employee authentication, attendance records, and shift management.
- Diagnosed and resolved 4 critical production bugs in the attendance flow, improving reliability and UX consistency.

Sahatk Falahy — E-Commerce App

Stack: Flutter · Dart · Riverpod · Odoo JSON-RPC · dio · SharedPreferences · Android · iOS

Available on: [Google Play](#) | [App Store](#)

- Full-featured e-commerce application for agricultural products, live on both platforms with 7,000+ downloads.
- Built and integrated Odoo-backed product catalog, cart, order management, and delivery pricing — all driven through custom JSON-RPC endpoints.
- Resolved complex bugs in address management and location/zone selection that were blocking the checkout flow.
- Owned the Android release pipeline: keystore setup, Gradle signing config (Kotlin DSL), and Google Play submission.

PERSONAL PROJECTS

Pulse — AI-Powered Emotional Chat Application Jul 2026 – Aug 2026

Stack: *Flutter · Dart · BLoC/Cubit · Clean Architecture · Firebase · Gemini AI · Hive · GetIt*

- Built a real-time Flutter chat application that analyzes and visualizes the emotional mood of messages.
- Developed a hybrid mood detection engine combining weighted Arabic/English keyword analysis, negation handling, caching, and Gemini AI fallback.
- Implemented Firebase Authentication, Firestore real-time messaging, FCM notifications, and an emotional timeline for conversation analytics.

Wavora — Offline-First Android Music Player Mar 2026

Stack: *Android · Jetpack Compose · Media3/ExoPlayer · Hilt · Room FTS4 · WorkManager · GitHub Actions*

- Production-grade offline-first music player backed by 252 automated unit/integration tests achieving 80%+ coverage.
- Engineered background playback service using Media3/ExoPlayer with reactive state propagation via Kotlin Coroutines and StateFlow.
- Resolved a critical MediaController async race condition, optimizing UI rendering performance by up to 40%.
- Automated CI/CD pipeline via GitHub Actions for build validation and test execution on every PR.

TECHNICAL SKILLS

Flutter: Dart · BLoC/Cubit · Riverpod · Dio · GetIt/Injectable · GoRouter · Firebase · Hive

Android: Kotlin · Jetpack Compose · Coroutines · Flow · Hilt · Room · WorkManager · Media3/ExoPlayer · Retrofit · Firebase

Backend & APIs: Python/Odoo ORM · JSON-RPC · REST APIs · Supabase

Architecture: Clean Architecture · MVVM · Repository Pattern · Modularization · SOLID

Testing & DevOps: JUnit · Mockito · BlocTest · GitHub Actions · Git · Gradle Kotlin DSL

Tools: Android Studio · VS Code · Postman · Firebase Console · GitHub

Languages: Arabic (Native) · English (Professional)

EDUCATION

B.Sc. Computer Science & Information Systems — [Ain Shams University, Cairo](#) Sep 2020 – Jul 2024

VOLUNTEERING

Android Developer — [MSP Student Activity](#) — [Ain Shams University](#) Nov 2021 – May 2022

- ❑ Built the Galaxy App (Java/MVVM) featuring Room Database, LiveData, and a full CRUD system with an interactive UI.